

TECHNICAL DOCUMENTATION

EngPro Production System

Technical Design Specification — Frontend · Backend · Database · Migration

React 18 + Nginx · NestJS 10 + Prisma · PostgreSQL 16

Version 1.0 · 6 August 2026 · Implements PRD v2.0 · Grounded in production data export (db/)

Document control

Item	Detail
Title	EngPro Production System — Technical Documentation v1.0
Implements	PRD v2.0 (Production Build Specification). Functional requirements, RBAC intent and user stories live in the PRD; this document defines HOW they are built.
Grounding data	Production CSV export in /db: workspaces (3 rows), members (23), site_admins (19), tickets (226, 18 columns), notes (337), kpi_settings (1), kpi_entries (146). Analysed 6 Aug 2026; findings in Section 2.
Audience	Backend and frontend engineers, DevOps, DBA, QA
Status	Draft for engineering review

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1. Introduction

1.1 Purpose

This document is the engineering blueprint for the EngPro production rebuild specified in PRD v2.0. It defines the database schema (DDL and data dictionary), the backend service design (NestJS modules, API contracts, transactions, realtime protocol), the frontend application design (React structure, data layer, components), the infrastructure (Docker, Nginx, CI/CD), and — critically — the data migration from the live legacy Supabase database, whose actual structure and content were analysed from the CSV export in the /db folder.

1.2 What the production data export changed

The /db export is the live database, and it corrects and extends several assumptions made from the source-code review alone:

- The live tickets table has 18 columns — including `tender_status` and `tender_value`, which do not exist in the reviewed application source. They confirm the deployed fork added tender-outcome tracking. Both columns are currently empty in all 225 production tickets, so the feature is schema-only today. The target schema adopts and formalises them (Section 4).
- Ticket numbering uses the EST- prefix in production (not EP- as shown in the deployed screenshot). The workspace prefix therefore becomes per-workspace configuration, defaulting to EST for the migrated workspace to preserve continuity (Decision D-1).
- The predicted duplicate-ticket-number defect is real and severe: 32 of 225 production tickets (14%) carry a `ticket_no` that at least one other ticket also carries — 20+ numbers are duplicated (EST-088, EST-126, EST-098, ...). Migration must repair this (Section 7.4).
- The KPI opening-points feature has never been used in production (146 entries: 139 auto, 7 manual, 0 opening). It remains in scope but is deprioritised in UAT emphasis.
- Two of the three workspaces are abandoned test workspaces ("KPN Downstream-EstimaX", "KPN Downstream-Estimation Control" [sic]); only workspace 73e3be57-... (KPN Downstream-Estimation Control) migrates.

1.3 Conventions

- All identifiers in code blocks are literal. SQL targets PostgreSQL 16. JSON examples are abridged with "..." where fields are obvious.
- MUST/SHOULD carry RFC-2119 meaning. Anything marked D-n is a recorded design decision; O-n is an open item.

2. Legacy database analysis (as-found)

2.1 Legacy schema

Table	Columns (as exported)	Rows
workspaces	id, company, subtitle, pw_manager, pw_admin, pw_member, created_at	3 (1 production + 2 test)
members	id, workspace_id, name, color, role, created_at	23 (12 in production: 8 member + 4 admin)
site_admins	id, workspace_id, site_name, password, color, created_at	19 (11 in production)
tickets	id, workspace_id, ticket_no, name, type, requestor, priority, status, assignee, deadline, source, description, created_at, updated_at, submitted_by, site_name, tender_status, tender_value	226 (225 in production)
notes	id, ticket_id, author, content, status_at_time, created_at	337
kpi_settings	id, workspace_id, point_on_target, point_miss_target, point_opening, created_at, updated_at	1 (3 / -2 / 10)
kpi_entries	id, workspace_id, member_name, month, year, points, type, description, ticket_id, ticket_no, created_by, created_at	146

2.2 Production data profile (workspace 73e3be57, 15 May – 4 Aug 2026)

Dimension	Observed values and distribution
Volume & velocity	225 tickets in ~12 weeks (~19/week); 337 notes; 146 KPI entries.
Status	Done 139 · New 42 · In Progress Tender 23 · In Progress Estimation 20 · Hold 1
Type	Budgeting Internal 65 · Project Tender 63 · Site Instruction / Site Memo 58 · Ops Tender 39
Priority	Normal 163 · Urgent 62 · Low 0 (Low is never used)
Source	Email 159 · WhatsApp 44 · Verbal / meeting 20 · empty 2
Deadline	Present on 100% of tickets (0 empty) — supports making deadline mandatory (PRD O-Q2).
Assignees	Rully 76 · Sajali 51 · Yohana 42 · Wahyu 32 · Danu 7 · Luqman 7 · Sumiardi 2 · Sumiardi_MGR 2 · Halomoan 1. Admin-role members do get assignments — the target model must allow ADMIN as assignee.
Intake channels	submitted_by is a member name for office intake (Seila 116, Faktul 28) and a site name for site intake (Dumai 41, Medan 21, Bontang 9, Karawang 5, Tj Pura 4, Gresik 1).
Field lengths	name max 179 chars; description max 533 (only 1 over 500); note content well under 2,000.
KPI	auto: 139 entries, values only +3 / -2 (matches settings); manual: 7 entries (-10, +2, +10); opening: never used. Months: May 40, Jun 54, Jul 51, Aug 1. No duplicate auto award per ticket (the race never fired — luck, not protection).
Notes authors	Named members plus the literal string "Manager" on 80 notes (the legacy app hard-codes author = "Manager" for manager sessions) — needs identity mapping in migration.
Referential health	Currently zero orphans: every assignee matches a member, every ticket site_name matches a site_admins row, every note points at an existing ticket. Name-cascade renames have held up so far.

2.3 Data-quality findings that drive design

#	Finding	Design consequence
DQ-1	32 of 225 tickets share duplicated ticket numbers (EST-088, EST-126, EST-098, EST-099, EST-023, EST-188, EST-080, EST-100, EST-093, EST-096, EST-016, EST-094, EST-081, EST-048, EST-092, EST-078, EST-097, EST-174, EST-095, EST-079, ...). Empirical proof of the count+1 race (PRD DATA-1).	Target schema enforces UNIQUE (workspace_id, ticket_no); migration rennumbers duplicates deterministically and preserves the original in legacy_ticket_no (Section 7.4).
DQ-2	tender_status and tender_value exist but are 100% empty.	Adopted into the target schema as first-class nullable fields with a proper enum + numeric type; UI exposure lands with the Tender-tracking feature (O-1).
DQ-3	Notes authored by the literal "Manager" (80 rows); kpi_entries.created_by is also a free-text label ("System"/"Manager").	Migration maps "Manager" to the seeded manager user; created_by becomes a nullable FK + kept text label for provenance.
DQ-4	Site name near-duplicates exist across workspaces (Karawang vs Kwarang, TJ Pura vs Tj Pura) — only test-workspace variants; production sites are clean but case-sensitive matching is fragile.	Target sites.name is citext with UNIQUE (workspace_id, name); migration report flags near-duplicates (Levenshtein ≤ 2) for manual confirmation.
DQ-5	Role passwords are 4-digit numeric strings (e.g. 2803, 4321, 1234); site passwords likewise.	Confirms SEC-3 urgency; no legacy password is imported; all users start with generated temporary passwords.
DQ-6	Two abandoned test workspaces hold junk data (member "Ggg", site "Lubug Gaung", etc.).	Migration scope is a single-workspace allowlist; everything else is archived to cold storage, not migrated.
DQ-7	Priority "Low" and KPI "opening" are defined but unused.	Both stay (they are cheap), but fixtures and UAT scripts focus on the used paths.

3. System architecture

3.1 Components and versions

Component	Technology	Version (pin)	Responsibility
Reverse proxy / static host	Nginx	1.27-alpine	TLS, security headers, SPA hosting, /api and /ws proxying, rate limiting
Frontend	React + TypeScript + Vite	React 18.3 / TS 5.x / Vite 6	SPA, role-scoped UI, optimistic updates
FE data layer	TanStack Query + socket.io-client	5.x / 4.x	Server-state cache + realtime invalidation
Backend	NestJS (Node 20 LTS)	10.x	REST API, RBAC, transactions, WebSocket gateway
ORM / migrations	Prisma	5.x	Typed data access, versioned schema migrations
Database	PostgreSQL	16-alpine	Source of truth, constraints, sequences
Auth	JWT (@nestjs/jwt) + argon2	HS512	Access/refresh tokens, password hashing
Containers	Docker Compose	v2	nginx + api + db + backup services
CI/CD	GitHub Actions	—	Lint, test, build, deploy

3.2 Runtime topology



3.3 Repository layout (monorepo)

```
engpro/
  apps/
    api/      # NestJS backend
    web/      # React frontend
  packages/
    shared/   # zod schemas + TS types shared FE/BE (enums, DTO shapes)
  infra/
    docker-compose.yml  docker-compose.prod.yml
    nginx/engpro.conf   Dockerfile.api  Dockerfile.web
  migration/           # legacy CSV -> Postgres ETL scripts + reports
  .github/workflows/ci.yml  deploy.yml
```

4. Database design

4.1 Complete DDL (authoritative, applied via Prisma migrations)

```
CREATE EXTENSION IF NOT EXISTS pg_trgm;
CREATE EXTENSION IF NOT EXISTS citext;

CREATE TYPE user_role AS ENUM ('MANAGER', 'ADMIN', 'SITE_ADMIN', 'ESTIMATOR');
CREATE TYPE ticket_status AS ENUM ('NEW', 'IN_PROGRESS_ESTIMATION',
    'IN_PROGRESS_TENDER', 'DONE', 'HOLD');
CREATE TYPE ticket_priority AS ENUM ('URGENT', 'NORMAL', 'LOW');
CREATE TYPE work_type AS ENUM ('PROJECT_TENDER', 'OPS_TENDER',
    'SITE_INSTRUCTION', 'BUDGETING_INTERNAL');
CREATE TYPE request_source AS ENUM ('WHATSAPP', 'EMAIL', 'VERBAL');
CREATE TYPE kpi_entry_type AS ENUM ('OPENING', 'AUTO', 'MANUAL', 'REVERSAL');
CREATE TYPE tender_status AS ENUM ('SUBMITTED', 'WON', 'LOST', 'CANCELLED');

CREATE TABLE workspaces (
    id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
    company text NOT NULL,
    subtitle text,
    ticket_prefix text NOT NULL DEFAULT 'EST',
    ticket_seq bigint NOT NULL DEFAULT 0,
    timezone text NOT NULL DEFAULT 'Asia/Jakarta',
    created_at timestamptz NOT NULL DEFAULT now(),
    updated_at timestamptz NOT NULL DEFAULT now()
);

CREATE TABLE sites (
    id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
    workspace_id uuid NOT NULL REFERENCES workspaces(id) ON DELETE CASCADE,
    name citext NOT NULL,
    color int NOT NULL DEFAULT 3,
    is_active boolean NOT NULL DEFAULT true,
    created_at timestamptz NOT NULL DEFAULT now(),
    UNIQUE (workspace_id, name)
);

CREATE TABLE users (
    id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
    workspace_id uuid NOT NULL REFERENCES workspaces(id) ON DELETE CASCADE,
    username citext NOT NULL,
    full_name text NOT NULL,
    email citext,
    password_hash text NOT NULL,
    role user_role NOT NULL,
    site_id uuid REFERENCES sites(id) ON DELETE RESTRICT,
    avatar_color int NOT NULL DEFAULT 0,
    is_active boolean NOT NULL DEFAULT true,
    must_change_password boolean NOT NULL DEFAULT true,
    created_at timestamptz NOT NULL DEFAULT now(),
    updated_at timestamptz NOT NULL DEFAULT now(),
    deleted_at timestamptz,
    UNIQUE (workspace_id, username),
    CHECK (role <> 'SITE_ADMIN' OR site_id IS NOT NULL)
);

CREATE TABLE tickets (
    id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
    workspace_id uuid NOT NULL REFERENCES workspaces(id) ON DELETE CASCADE,
    ticket_no text NOT NULL,
    legacy_ticket_no text, -- pre-migration number if renumbered
    name text NOT NULL CHECK (char_length(name) BETWEEN 3 AND 200),
    type work_type NOT NULL,
    priority ticket_priority NOT NULL,
    status ticket_status NOT NULL DEFAULT 'NEW',
```

```

requestor      text,
source         request_source,
description    text CHECK (char_length(description) <= 5000),
deadline      date NOT NULL,          -- D-2: mandatory (100% populated in legacy)
assignee_id    uuid REFERENCES users(id) ON DELETE SET NULL,
site_id       uuid REFERENCES sites(id) ON DELETE SET NULL,
submitted_by_id uuid REFERENCES users(id) ON DELETE SET NULL,
tender_status  tender_status,        -- fork columns, formalised (DQ-2)
tender_value   numeric(18,2) CHECK (tender_value IS NULL OR tender_value >= 0),
version       int NOT NULL DEFAULT 1,
created_at    timestampz NOT NULL DEFAULT now(),
updated_at    timestampz NOT NULL DEFAULT now(),
completed_at   timestampz,
deleted_at    timestampz,
UNIQUE (workspace_id, ticket_no),
CHECK (tender_status IS NULL OR type IN ('PROJECT_TENDER','OPS_TENDER'))
);

CREATE TABLE ticket_notes (
  id          uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  ticket_id   uuid NOT NULL REFERENCES tickets(id) ON DELETE CASCADE,
  author_id   uuid REFERENCES users(id) ON DELETE SET NULL,
  author_label text NOT NULL,          -- denormalised display name (provenance)
  content     text NOT NULL CHECK (char_length(content) BETWEEN 1 AND 2000),
  status_at_time ticket_status NOT NULL,
  created_at  timestampz NOT NULL DEFAULT now()
);

CREATE TABLE kpi_settings (
  workspace_id  uuid PRIMARY KEY REFERENCES workspaces(id) ON DELETE CASCADE,
  point_opening int NOT NULL DEFAULT 10,
  point_on_target int NOT NULL DEFAULT 3,
  point_miss_target int NOT NULL DEFAULT -2,
  updated_by_id uuid REFERENCES users(id),
  updated_at    timestampz NOT NULL DEFAULT now()
);

CREATE TABLE kpi_entries (
  id          uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  workspace_id uuid NOT NULL REFERENCES workspaces(id) ON DELETE CASCADE,
  user_id     uuid NOT NULL REFERENCES users(id) ON DELETE RESTRICT,
  month       int NOT NULL CHECK (month BETWEEN 1 AND 12),
  year        int NOT NULL CHECK (year BETWEEN 2020 AND 2100),
  points      int NOT NULL,
  type        kpi_entry_type NOT NULL,
  description  text,
  ticket_id   uuid REFERENCES tickets(id) ON DELETE SET NULL,
  ticket_no   text,                  -- survives ticket deletion
  created_by_id uuid REFERENCES users(id),
  created_by  text NOT NULL DEFAULT 'System',
  created_at  timestampz NOT NULL DEFAULT now()
);
CREATE UNIQUE INDEX ux_kpi_auto_once ON kpi_entries (ticket_id) WHERE type = 'AUTO';
CREATE UNIQUE INDEX ux_kpi_opening  ON kpi_entries (user_id, year, month)
  WHERE type = 'OPENING';

CREATE TABLE audit_log (
  id          bigserial PRIMARY KEY,
  workspace_id uuid NOT NULL,
  actor_id    uuid REFERENCES users(id),
  entity_type text NOT NULL,        -- ticket | user | site | kpi_entry | ...
  entity_id   uuid,
  action      text NOT NULL,        -- create | update | delete | restore | login | ...
  before      jsonb,
  after       jsonb,
  ip          inet,
  created_at  timestampz NOT NULL DEFAULT now()
);

```

```

CREATE TABLE refresh_tokens (
  id          uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id     uuid NOT NULL REFERENCES users(id) ON DELETE CASCADE,
  family_id   uuid NOT NULL,          -- rotation chain; replay revokes the family
  token_hash  text NOT NULL,
  user_agent  text,
  expires_at  timestampz NOT NULL,
  revoked_at  timestampz,
  created_at  timestampz NOT NULL DEFAULT now()
);

-- performance indexes
CREATE INDEX ix_tickets_ws_status  ON tickets (workspace_id, status)
  WHERE deleted_at IS NULL;
CREATE INDEX ix_tickets_ws_assignee ON tickets (workspace_id, assignee_id)
  WHERE deleted_at IS NULL;
CREATE INDEX ix_tickets_ws_site    ON tickets (workspace_id, site_id)
  WHERE deleted_at IS NULL;
CREATE INDEX ix_tickets_ws_deadline ON tickets (workspace_id, deadline)
  WHERE deleted_at IS NULL;
CREATE INDEX ix_tickets_name_trgm  ON tickets USING gin (name gin_trgm_ops);
CREATE INDEX ix_notes_ticket       ON ticket_notes (ticket_id, created_at);
CREATE INDEX ix_kpi_ws_period      ON kpi_entries (workspace_id, year, month);
CREATE INDEX ix_kpi_user_year      ON kpi_entries (user_id, year);
CREATE INDEX ix_audit_entity       ON audit_log (entity_type, entity_id);

```

4.2 Data dictionary — tickets (with legacy mapping)

Column	Type	Null	Constraint / default	Legacy source (db/*.csv)
id	uuid	N	PK, gen_random_uuid()	tickets.id (preserved verbatim)
workspace_id	uuid	N	FK workspaces	tickets.workspace_id
ticket_no	text	N	UNIQUE (workspace_id, ticket_no)	tickets.ticket_no; duplicates renumbered (7.4)
legacy_ticket_no	text	Y	set only when renumbered	= original tickets.ticket_no
name	text	N	len 3–200 (legacy max 179)	tickets.name
type	work_type	N	enum	tickets.type via label map (2.2)
priority	ticket_priority	N	enum	tickets.priority (upper-cased)
status	ticket_status	N	enum, default NEW	tickets.status via label map
requestor	text	Y	len ≤ 120	tickets.requestor
source	request_source	Y	enum; NULL for the 2 legacy blanks	tickets.source
description	text	Y	len ≤ 5000 (legacy max 533)	tickets.description
deadline	date	N	mandatory (D-2)	tickets.deadline (100% populated)
assignee_id	uuid	Y	FK users, SET NULL	lookup users by tickets.assignee name
site_id	uuid	Y	FK sites, SET NULL	lookup sites by tickets.site_name
submitted_by_id	uuid	Y	FK users, SET NULL	tickets.submitted_by → user (member name) or the site's SITE_ADMIN user (site name)

Column	Type	Null	Constraint / default	Legacy source (db/*.csv)
tender_status	tender_status	Y	enum; only for tender types (CHECK)	tickets.tender_status (all empty today)
tender_value	numeric(18,2)	Y	≥ 0	tickets.tender_value (all empty today)
version	int	N	default 1; optimistic lock	— (new)
created_at / updated_at	timestampz	N	defaults now()	tickets.created_at / updated_at
completed_at	timestampz	Y	set on DONE transition	= updated_at when legacy status = Done
deleted_at	timestampz	Y	soft delete	— (new)

4.3 Data dictionary — users (merges legacy members + site_admins)

Column	Type	Null	Constraint / default	Legacy source (db/*.csv)
id	uuid	N	PK	members.id preserved; site-admin users get new ids
username	citext	N	UNIQUE per workspace	slug of members.name / site_admins.site_name (e.g. rully, sumiardi_mgr, site.dumai)
full_name	text	N	—	members.name / "<site> Site Admin"
role	user_role	N	enum	members.role: admin→ADMIN, member→ESTIMATOR; site_admins→SITE_ADMIN; seeded manager→MANAGER
site_id	uuid	Y	required when role = SITE_ADMIN (CHECK)	site_admins.id → sites.id
password_hash	text	N	argon2id	NEVER migrated (DQ-5); temp passwords issued
avatar_color	int	N	0–8 palette	members.color / site_admins.color
must_change_password	bool	N	default true	— (new; true for all migrated users)

4.4 Data dictionary — kpi_entries

Column	Type	Null	Constraint / default	Legacy source (db/*.csv)
user_id	uuid	N	FK users RESTRICT	kpi_entries.member_name → user lookup
month / year	int	N	CHECK ranges	kpi_entries.month / year
points	int	N	—	kpi_entries.points (observed: +3/–2 auto; –10/+2/+10 manual)

Column	Type	Null	Constraint / default	Legacy source (db/*.csv)
type	kpi_entry_type	N	enum + partial unique indexes	kpi_entries.type: auto→AUTO, manual→MANUAL ('opening' unused in legacy)
ticket_id / ticket_no	uuid / text	Y	FK SET NULL / label survives deletion	kpi_entries.ticket_id / ticket_no
created_by_id + created_by	uuid + text	Y/N	FK + label	created_by "System"→NULL+label; "Manager"→manager user

workspaces, sites, ticket_notes, kpi_settings, audit_log and refresh_tokens follow the DDL in 4.1; their legacy mappings are one-to-one and covered in Section 7.

5. Backend design (NestJS)

5.1 Project structure

```
apps/api/src/
  main.ts                # helmet, CORS, validation pipe, swagger, shutdown hooks
  app.module.ts
  prisma/                # PrismaService (tx helpers), schema.prisma
  common/
    guards/ jwt-auth.guard.ts roles.guard.ts password-change.guard.ts
    decorators/ roles.decorator.ts current-user.decorator.ts
    interceptors/ audit.interceptor.ts
    filters/ http-exception.filter.ts # RFC-7807 error shape
  auth/ auth.controller.ts auth.service.ts token.service.ts dto/
  users/ users.controller.ts users.service.ts dto/
  sites/ ...
  workspace/ ...
  tickets/ tickets.controller.ts tickets.service.ts ticket-number.service.ts
           permission-envelope.ts dto/
  notes/ ...
  kpi/ kpi.controller.ts kpi.service.ts kpi-award.service.ts dto/
  audit/ ...
  events/ events.gateway.ts events.service.ts
  health/ health.controller.ts
```

5.2 Authentication internals

Aspect	Specification
Access token	JWT HS512, TTL 15 min. Payload: { sub: userId, ws: workspaceId, role, siteId?, mcp?: mustChangePassword, iat, exp }. Sent as Authorization: Bearer.
Refresh token	256-bit random, TTL 7 d, stored as SHA-256 hash in refresh_tokens with family_id. Delivered as httpOnly Secure SameSite=Strict cookie scoped to /api/v1/auth. On refresh: verify hash + not revoked + not expired → issue new pair, revoke old row (same family). Reuse of a revoked token → revoke entire family, audit "token-replay".
Password hashing	argon2id: memory 64 MiB, time 3, parallelism 1 (~100 ms on target host). Policy: min 10 chars, denylist top-10k, max 128.
Login throttling	@nestjs/throttler: 5/min per IP on /auth/login + progressive per-username delay (2^n s, cap 60 s) tracked in memory (Redis if scaled out). Uniform 401 body prevents user enumeration.
Guard pipeline	JwtAuthGuard → PasswordChangeGuard (blocks all but /auth/change-password when mcp) → RolesGuard (@Roles metadata) → service-level scoping (workspace_id from token; SITE_ADMIN additionally site_id).

5.3 API contracts — representative examples

5.3.1 POST /api/v1/auth/login

```
Request { "username": "rully", "password": "*****" }
Response 200
{
  "accessToken": "eyJhbGciOiJIUzUxMiIs... ",
  "user": { "id": "36afd736-...", "username": "rully", "fullName": "Rully",
            "role": "ESTIMATOR", "avatarColor": 0, "mustChangePassword": false },
  "workspace": { "company": "KPN Downstream-Estimation Control",
                 "ticketPrefix": "EST" }
}
Set-Cookie: engpro_rt=...; HttpOnly; Secure; SameSite=Strict; Path=/api/v1/auth
Errors: 401 invalid credentials (uniform), 429 throttled
```

5.3.2 POST /api/v1/tickets

```
Request
{ "name": "ADD 1000MT SODIUM STORAGE TANK - BONTANG (EI Package)",
  "type": "PROJECT_TENDER", "priority": "NORMAL", "deadline": "2026-09-19",
  "requestor": "PT KPN", "source": "EMAIL",
  "description": "OSBL piping + EI packages", "assigneeId": null, }
Response 201
{ "id": "...", "ticketNo": "EST-226", "status": "NEW", "version": 1, "createdAt": "..." }
Server behaviour: assigneeId accepted only for MANAGER/ADMIN; SITE_ADMIN requests
force siteId + submittedById from the JWT and ignore client-sent values.
Errors: 400 validation (field list), 403 role
```

5.3.3 PATCH /api/v1/tickets/:id (status change with KPI side-effect)

```
Request { "status": "DONE", "version": 4 }
Response 200
{ "id": "...", "status": "DONE", "version": 5, "completedAt": "2026-08-06T09:12:00Z",
  "kpiAward": { "type": "AUTO", "points": 3, "reason": "ON_TARGET" } }
Errors:
409 { "error": "VersionConflict", "currentVersion": 6 } -- stale client
403 { "error": "FieldNotPermitted", "fields": ["assigneeId"] }
```

5.3.4 GET /api/v1/tickets (list)

```
GET /api/v1/tickets?page=1&pageSize=25&status=NEW&search=sodium
&assigneeId=unassigned&sort=-createdAt
Response 200
{ "items": [ { "id": "...", "ticketNo": "EST-224", "name": "...", "type": "PROJECT_TENDER",
  "priority": "NORMAL", "status": "NEW", "deadline": "2026-09-19", "isOverdue": false,
  "assignee": null, "site": { "id": "...", "name": "Bontang" },
  "lastNote": { "content": "...", "authorLabel": "Sajali", "createdAt": "..." },
  "noteCount": 3, "version": 2 } ],
  "total": 42, "page": 1, "pageSize": 25 }
```

5.3.5 GET /api/v1/tickets/stats (dashboard)

```
{ "total": 225, "done": 139, "completionRate": 0.62, "overdue": 7, "unassigned": 10,
  "byStatus": { "NEW": 42, "IN_PROGRESS_ESTIMATION": 20, "IN_PROGRESS_TENDER": 23,
    "DONE": 139, "HOLD": 1 },
  "byType": { "PROJECT_TENDER": 63, "OPS_TENDER": 39, "SITE_INSTRUCTION": 58,
    "BUDGETING_INTERNAL": 65 },
  "workload": [ { "userId": "...", "fullName": "Rully", "active": 15, "done": 61 }, ... ] }
```

5.4 Transactional flows

5.4.1 Ticket creation (race-free numbering)

```
prisma.$transaction(async (tx) => {
  const ws = await tx.$queryRaw`
    UPDATE workspaces SET ticket_seq = ticket_seq + 1
    WHERE id = ${wsId} RETURNING ticket_seq, ticket_prefix`;
  const n = ws[0].ticket_seq;
  const ticketNo = `${ws[0].ticket_prefix}-${String(n).padStart(3, "0")}`;
  const ticket = await tx.ticket.create({ data: { ...dto, ticketNo, ... } });
  await audit(tx, actor, "ticket", ticket.id, "create", null, ticket);
  return ticket;
}); // then eventsService.emit("ticket.created", { id, workspaceId })
```

5.4.2 Status change with KPI award / reversal (single transaction)

```
prisma.$transaction(async (tx) => {
  const upd = await tx.$executeRaw`
    UPDATE tickets SET status=${s}, version=version+1, updated_at=now(),
    completed_at = CASE WHEN ${s}='DONE' THEN now() ELSE NULL END
    WHERE id=${id} AND version=${dto.version} AND deleted_at IS NULL`;
  if (upd === 0) throw new ConflictException(); // 409
  if (s === "DONE" && prev !== "DONE" && assigneeId) {
    const onTime = todayInWsTz() <= ticket.deadline;
    try {
      await tx.kpiEntry.create({ data: { type: "AUTO",
```

```

        points: onTime ? cfg.pointOnTarget : cfg.pointMissTarget, ... } }));
    } catch (e) { if (!isUniqueViolation(e)) throw e; } // already awarded: OK
  }
  if (prev === "DONE" && s !== "DONE") { // reversal
    const auto = await sumAutoAndReversals(tx, id);
    if (auto !== 0) await tx.kpiEntry.create({ data: { type: "REVERSAL",
      points: -auto, description: `Reversal - reopened ${ticketNo}` } });
  }
  await audit(tx, actor, "ticket", id, "update", before, after);
});

```

5.5 WebSocket protocol (socket.io, namespace /ws)

Event	Payload	Emitted when
ticket.created	{ id, workspaceId, siteId? }	After create commit
ticket.updated	{ id, workspaceId, changed: ["status","assigneeId",...] }	After update commit
ticket.deleted / ticket.restored	{ id, workspaceId }	Soft delete / restore
note.created	{ ticketId, workspaceId }	After note commit
kpi.updated	{ workspaceId, userId, year, month }	Any KPI entry written
user.updated	{ id, workspaceId, deactivated? }	User admin changes; deactivated=true forces client logout

- Handshake: auth: { token: <accessJWT> }; invalid/expired → disconnect with reason "unauthorized"; client refreshes and reconnects.
- Rooms: every socket joins ws:<workspaceId>; SITE_ADMIN sockets get site-filtered ticket events (server filters before emit to the site room site:<siteId>).
- Clients treat events as cache-invalidation hints only — they refetch via REST; payloads never carry row data (keeps authz single-pathed).

5.6 Validation & error model

Field	Rule (informed by legacy profile)
ticket.name	string 3–200 (legacy max observed 179)
ticket.description	≤ 5000 (legacy max 533)
ticket.requestor	≤ 120
ticket.deadline	ISO date, required; past dates require allowPast: true flag (intake of backdated requests)
note.content	1–2000
kpi manual points	integer, ≠ 0, points ≤ 100; description required 3–200
username	^[a-z0-9._-]{3,32}\$ (citext-unique per workspace)
password	10–128, denylist checked

Error envelope (all non-2xx):

```

{ "statusCode": 409, "error": "VersionConflict",
  "message": "Ticket was modified by another user.",
  "details": { "currentVersion": 6 }, "traceId": "req-8f3a..." }

```

6. Frontend design (React + Vite)

6.1 Project structure

```
apps/web/src/
  main.tsx App.tsx router.tsx
  api/      client.ts (fetch wrapper: auth header, 401->refresh->retry once)
            endpoints/ tickets.ts users.ts kpi.ts ... (typed via packages/shared)
  auth/     AuthProvider.tsx useAuth() RequireRole.tsx ChangePasswordGate.tsx
  realtime/ SocketProvider.tsx useInvalidateOnEvent.ts
  features/
    dashboard/ StatsGrid StatusGrid TypeGrid WorkloadGrid RecentTickets
    board/      BoardPage BoardColumn TicketCard BoardFilters useDragMove
    tickets/    ListPage TicketTable Filters SearchBox Pagination
    ticket/     DetailPage MetaPanel EditForm NotesTimeline DeleteDialog
    intake/     NewTicketPage (office + site variants)
    myTickets/  MyTicketsPage OverdueBanner UrgentSection
    kpi/        ManagerKpiPage MonthStrip LeaderboardCard LedgerTable
                SettingsModal OpeningModal BonusModal MemberKpiPage
    settings/   UsersTab SitesTab WorkspaceTab AuditTab
  components/ Badge Avatar Modal Toast EmptyState SyncDot DataTable
  i18n/        en.json id.json
  styles/      tokens.css (design tokens carried from the validated v1 UI)
```

6.2 Data layer conventions

Concern	Convention
Query keys	['tickets', filters] · ['ticket', id] · ['tickets','stats'] · ['kpi','summary',year,month] · ['users'] · ['audit',filters]
Realtime invalidation	ticket.* → invalidate ['tickets'] + ['ticket',id] + ['tickets','stats']; note.created → ['ticket',ticketId]; kpi.updated → ['kpi',*]. Debounce 500 ms to coalesce bursts.
Optimistic updates	Board drag: setQueryData moves the card immediately; onError restores snapshot and toasts; onSettled invalidates. 409 handling refetches and re-applies nothing.
Auth storage	Access token in memory (module scope); never localStorage. Refresh via cookie. On hard reload: silent refresh before first render (splash ≤ 300 ms).
Token in socket	SocketProvider re-authenticates the socket whenever a new access token is issued.
Escaping	JSX only; dangerouslySetInnerHTML banned by ESLint rule (closes legacy XSS class).

6.3 Role-based UI composition

Role	Navigation (from JWT role claim)	Guard behaviour
MANAGER	Dashboard · Board · Tickets · KPI · Settings	RequireRole redirects others to their home
ADMIN	Dashboard · Board · Tickets · New ticket	No /settings, /kpi routes
SITE_ADMIN	New request · My site tickets	All ticket queries carry the server-enforced site scope; UI shows site chip
ESTIMATOR	My tickets · Team board · My KPI	Board cards draggable only when assigneeld = me

6.4 Design tokens (carried from validated legacy UI)

```
:root {
  --bg:#F4F3EF; --surface:#FFF; --text:#181816; --muted:#6B6A66;
  --accent:#2D5BE3; --accent-bg:#EBF0FD; --red:#C0392B; --red-bg:#FDECEB;
  --green-bg:#E8F5E9; --amber-bg:#FDF4E3; --purple-bg:#F0EEFF; --teal-bg:#E0F7F2;
```

```
--avatar-palette: #2D5BE3 #8B5CF6 #0D9488 #D97706 #DC2626
                  #059669 #7C3AED #C026D3 #0369A1; /* indexes 0-8 */
}
Status badge mapping: NEW=accent IN_PROGRESS_ESTIMATION=amber
IN_PROGRESS_TENDER=purple DONE=green HOLD=gray
Type badge mapping: PROJECT_TENDER=purple OPS_TENDER=teal
SITE_INSTRUCTION=amber BUDGETING_INTERNAL=gray
```

7. Data migration specification

7.1 Scope and principles

- Source: the /db CSV export re-taken at cutover (fresh export, same shapes). Target: PostgreSQL 16 schema of Section 4.
- Scope allowlist: workspace 73e3be57-c404-4463-b336-8d429e022477 only. The two test workspaces (77b4ff3f-..., d250c87e-...) are archived as raw CSV in cold storage and NOT migrated (DQ-6).
- Idempotent ETL: every script runs against a clean target and is safe to re-run (truncate-and-load per table); rehearsed on staging until the validation gate passes twice consecutively.
- Everything the ETL changes is written to migration/reports/*.csv for sign-off (renumbered tickets, identity mappings, skipped rows).

7.2 Identity mapping (users)

1. Seed the MANAGER account (username manager, full name from operations input); it also absorbs legacy authorship labels "Manager" (80 notes) and kpi_entries.created_by = "Manager".
2. members (12 production rows) → users: role admin→ADMIN (Sumiardi_MGR, Faktul, Seila, Rull_Y), member→ESTIMATOR (Wahyu, Luqman, Halomoan, Danu, Rully, Sumiardi, Yohana, Sajali). username = lowercased name with _ preserved; member ids preserved as user ids.
3. site_admins (11 production rows: Medan, Gresik, Karawang, Bontang, Tangerang, Bagendang, TPG, GLM, Dumai, Bekasi, Tj Pura) → one sites row each (id preserved) + one SITE_ADMIN user each (new id, username site.<slug>, site_id set).
4. All users: is_active = true, must_change_password = true, argon2id hash of a generated 12-char temporary password; the temp-password sheet is delivered to the manager out-of-band and purged after first-login confirmation.
5. Note: "Sumiardi" (ESTIMATOR) and "Sumiardi_MGR" (ADMIN) are intentionally distinct accounts — both hold ticket assignments in legacy data (2 each); do not merge.

7.3 Per-table mapping

Legacy → target	Transform rules
workspaces → workspaces	Company, subtitle, created_at copied; ticket_prefix = "EST" (D-1); ticket_seq = highest numeric suffix after renumbering (7.4); pw_* columns dropped.
members → users	Section 7.2. color → avatar_color.
site_admins → sites (+ users)	site_name → name (citext); password dropped; color kept.
tickets → tickets	Ids preserved. Label maps: status "In Progress Estimation"→IN_PROGRESS_ESTIMATION etc.; type "Site Instruction / Site Memo"→SITE_INSTRUCTION; priority upper-case; source "Verbal / meeting"→VERBAL, empty→NULL. assignee → assignee_id (name lookup); site_name → site_id; submitted_by → submitted_by_id (member name → that user; site name → the site's SITE_ADMIN user). completed_at = updated_at where status DONE. tender_status/tender_value: NULL (empty in legacy). version = 1.
notes → ticket_notes	author "Manager" → manager user; other names → user lookup; unmatched → author_id NULL. author_label = original string always (provenance). status_at_time label-mapped.

Legacy → target	Transform rules
kpi_settings → kpi_settings	One row (3 / -2 / 10) keyed by workspace_id.
kpi_entries → kpi_entries	member_name → user_id (all 8 names match users); type auto→AUTO, manual→MANUAL; created_by "System"→(NULL, label "System"), "Manager"→(manager id, label "Manager"); ticket refs preserved.

7.4 Duplicate ticket-number repair (DQ-1)

1. Group production tickets by ticket_no; 20+ numbers have 2+ holders (32 tickets affected of 225).
2. Within each duplicate group, the ticket with the EARLIEST created_at keeps the number.
3. Every later holder is renumbered to the next free number continuing the series (after the current maximum), in global created_at order — deterministic and re-runnable.
4. Renumbered tickets store their original number in legacy_ticket_no, and the ETL appends a system note: "Ticket renumbered from EST-xxx to EST-yyy during migration (duplicate repair)".
kpi_entries.ticket_no labels for renumbered tickets are updated to the new number (the ticket_id link is unaffected).
5. Report renumbered_tickets.csv (old number, new number, name, created_at) goes to the estimation team BEFORE cutover so site/office references can be reconciled.
6. Set workspaces.ticket_seq = numeric suffix of the highest assigned number, so the first post-migration ticket continues the series with no gap or collision.

7.5 Validation gate (must pass before cutover sign-off)

Check	Rule
Row counts	users = 12 members + 11 site admins + 1 manager = 24; sites = 11; tickets = 225; ticket_notes = 337; kpi_entries = 146; kpi_settings = 1.
Uniqueness	SELECT ticket_no, count(*) ... HAVING count(*) > 1 returns 0 rows; renumbered count = expected from report.
Referential	0 tickets with non-null assignee that fails the FK; 0 notes without ticket; 0 kpi_entries without user.
KPI reconciliation	Per member per month, SUM(points) identical between legacy CSV and target DB (8 members × 4 months).
Enum coverage	0 rows with unmapped status/type/priority/source labels.
Spot checks	10 random tickets compared field-by-field incl. note ordering and last-note excerpt.

7.6 Cutover runbook

1. T-7d: rehearse ETL on staging with a fresh export; distribute renumbered_tickets.csv and temp-password sheet drafts.
2. T-0 18:00 WIB: announce freeze; set legacy Supabase RLS policies to read-only (replace allow_all with USING (true) for SELECT only) — the legacy app keeps working read-only.
3. Export fresh CSVs; run ETL against production PostgreSQL; run validation gate.
4. Smoke test golden paths on production URL with pilot users (manager + 1 estimator + 1 site admin).
5. Switch DNS/bookmark to the new URL; deliver temp passwords; hypercare channel opens.
6. T+14d: decommission legacy Netlify site; final Supabase snapshot to cold storage; revoke anon key.

8. Infrastructure & operations

8.1 docker-compose.prod.yml (abridged)

```
services:
  nginx:
    image: nginx:1.27-alpine
    ports: ["80:80", "443:443"]
    volumes:
      - ./nginx/engpro.conf:/etc/nginx/conf.d/default.conf:ro
      - web-dist:/usr/share/nginx/html:ro
      - certs:/etc/letsencrypt:ro
    depends_on: [api]
  api:
    image: ghcr.io/<org>/engpro-api:${TAG}
    env_file: .env.production      # DATABASE_URL, JWT_*, CORS_ORIGIN, TZ
    depends_on: { db: { condition: service_healthy } }
    restart: unless-stopped
  db:
    image: postgres:16-alpine
    environment: { POSTGRES_DB: engpro, POSTGRES_USER: engpro }
    volumes: [pgdata:/var/lib/postgresql/data]
    healthcheck: { test: ["CMD-SHELL", "pg_isready -U engpro"], interval: 10s }
  backup:
    image: postgres:16-alpine
    entrypoint: ["/bin/sh", "-c",
      "while true; do pg_dump -h db -U engpro -Fc engpro \
        > /backups/engpro-$(date +%F).dump; find /backups -mtime +30 -delete; \
        sleep 86400; done"]
    volumes: [backups:/backups]
```

8.2 Nginx configuration (key directives)

```
limit_req_zone $binary_remote_addr zone=auth:10m rate=5r/m;
server {
  listen 443 ssl http2; server_name engpro.example.com;
  add_header Strict-Transport-Security "max-age=31536000" always;
  add_header X-Content-Type-Options nosniff always;
  add_header Content-Security-Policy "default-src 'self'; connect-src 'self' wss;;
    img-src 'self' data;; style-src 'self' 'unsafe-inline'" always;
  client_max_body_size 2m;
  location / { root /usr/share/nginx/html; try_files $uri /index.html;
    location ~* \.(js|css|woff2)$ { expires 1y; add_header Cache-Control immutable; } }
  location /api/v1/auth/ { limit_req zone=auth burst=5 nodelay;
    proxy_pass http://api:3000; }
  location /api/ { proxy_pass http://api:3000; proxy_set_header X-Real-IP $remote_addr; }
  location /ws/ { proxy_pass http://api:3000; proxy_http_version 1.1;
    proxy_set_header Upgrade $http_upgrade; proxy_set_header Connection "upgrade"; }
}
```

8.3 Environment variables

Variable	Example / rule
DATABASE_URL	postgresql://engpro:***@db:5432/engpro?connection_limit=20
JWT_ACCESS_SECRET / JWT_REFRESH_SECRET	≥ 64 random bytes each; rotated on incident; never equal
ACCESS_TTL / REFRESH_TTL	15m / 7d
CORS_ORIGIN	https://engpro.example.com (exact, no wildcard)
TZ / WORKSPACE_TZ default	Asia/Jakarta (deadline comparisons)
LOG_LEVEL	info (debug on staging)

8.4 CI/CD pipeline

1. PR: eslint (incl. no-dangerouslySetInnerHTML rule) → tsc → unit tests → API e2e vs postgres:16 service container → web component tests → gitleaks + npm audit (high fails).
2. main merge: build web dist + api image (multi-stage, node:20-alpine, non-root user) → tag :sha → push GHCR.
3. Staging auto-deploy: prisma migrate deploy → container swap → Playwright smoke (login, create ticket, drag card, KPI view).
4. Production: manual approval → same artefacts → migrate deploy → swap → smoke. Rollback = previous tag; migrations must be backward-compatible one release.

8.5 Observability & maintenance

- pino JSON logs with traceId; request logging excludes bodies on /auth/*. Alerts: 5xx rate > 1%/5min, failed logins > 20/5min, DB pool > 80%, disk > 80%, backup age > 26 h.
- Metrics endpoint /metrics (Prometheus format): http_request_duration histogram, socket_connections gauge, tickets_created_total counter.
- Quarterly restore drill from the newest dump into a scratch container; result logged in the ops journal.

9. Testing specification

9.1 Fixtures mirror production shape

Seed fixtures replicate the real profile from Section 2.2: 12 users (+manager), 11 sites, 225 tickets with the observed status/type distribution, 32 engineered duplicate legacy numbers for migration tests, 337 notes, 146 KPI entries across 4 months. This keeps performance and pagination tests honest.

9.2 Test matrix highlights

Area	Cases (must pass in CI)
Numbering	50 parallel POST /tickets → 50 unique consecutive numbers; sequence continues after soft delete + restore; prefix change affects only new tickets.
RBAC	Endpoint × role grid from PRD §4.3 (about 120 assertions, generated from a table); field-level PATCH denial per role; SITE_ADMIN cross-site read returns 404 (not 403 — no existence leak).
Optimistic locking	Two clients read v4; both PATCH; second gets 409 with currentVersion; audit shows one update.
KPI	On-time DONE → +3; late DONE → -2; deadline edited after DONE does not rescore; reopen → reversal; re-complete → fresh award; parallel double-DONE → single AUTO row (unique index).
Auth	Refresh rotation; replay revokes family; deactivation kills session ≤ access TTL; throttle after 5 failures; uniform 401 body.
Migration (ETL)	Validation gate of \$7.5 automated; duplicate repair determinism (same input → same output twice); KPI sums reconcile.
Realtime	Two socket clients: create/update/delete propagate < 1 s median; site room isolation (site A never receives site B events).
Load (k6)	List P95 < 300 ms @ 5k tickets / 30 VUs; stats P95 < 300 ms; login P95 < 400 ms (argon2 budget).

10. Decisions and open items

ID	Type	Item
D-1	Decision	Ticket prefix stays EST for the migrated workspace (production reality; the EP- prefix seen in one screenshot belongs to a different/testing dataset). Prefix is per-workspace config.
D-2	Decision	deadline becomes NOT NULL: 100% of legacy tickets carry one, and it powers overdue + KPI logic. Intake UI requires it (closes PRD open question Q2).
D-3	Decision	tender_status / tender_value are formalised in schema now (enum SUBMITTED/WON/LOST/CANCELLED + numeric(18,2)) but surfaced in UI only when the Tender-tracking feature ships (Phase 2). CHECK restricts them to tender-type tickets.
D-4	Decision	"Sumiardi" and "Sumiardi_MGR" migrate as separate accounts (both hold assignments).
D-5	Decision	Note authorship: author_id FK plus immutable author_label text — history keeps its original attribution even if users are renamed or deleted.
O-1	Open	Tender-tracking UX (where WON/LOST is set, whether tender_value is visible to estimators) — needs the deployed fork's source or product decision before Phase 2.
O-2	Open	Email delivery provider for temp passwords and Phase-2 notifications (SMTP relay vs transactional API).

ID	Type	Item
O-3	Open	Hosting target (customer VPS vs cloud VM) — affects certificate automation and backup off-siting only.

Appendix A — production data profile snapshot (6 Aug 2026)

Metric	Value
Workspaces in export	3 (1 production: "KPN Downstream-Estimation Control"; 2 test — excluded)
Production members	12 (8 ESTIMATOR: Wahyu, Luqman, Halomoan, Danu, Rully, Sumiardi, Yohana, Sajali · 4 ADMIN: Sumiardi_MGR, Faktul, Seila, Rull_Y)
Production sites	11 (Medan, Gresik, Karawang, Bontang, Tangerang, Bagendang, TPG, GLM, Dumai, Bekasi, Tj Pura)
Tickets	225 (15 May – 4 Aug 2026, ~19/week); 139 Done (62%), 42 New, 23 In Progress Tender, 20 In Progress Estimation, 1 Hold
Duplicate ticket numbers	32 tickets across 20+ numbers (14% of register) — repaired in migration §7.4
Site-submitted tickets	81 (Dumai 41, Medan 21, Bontang 9, Karawang 5, Tj Pura 4, Gresik 1); office intake: Seila 116, Faktul 28
Notes	337; authors incl. literal "Manager" ×80
KPI entries	146 = 139 AUTO (+3/–2 only) + 7 MANUAL (–10, +2, +10); opening never used; no duplicate AUTO per ticket
Field maxima	ticket name 179 chars; description 533; both far inside target limits
Unused vocabulary	Priority "Low": 0 uses; tender_status/tender_value: empty on all rows